CONTINUED

Discussion questions

1 What kind of feature would you include in a smart refrigerator that you would find helpful?

2 Can you see any problems that might arise as a result of connecting devices and the development of the Internet of Things?

3 Work with a partner and design your own smart device. What would you want it to do? What computer hardware would it need?

KEY WORDS

central processing unit (CPU): a component in a computer system that processes data and instructions.

microprocessor: an integrated circuit that is able to perform many of the functions of a CPU.

embedded system: a computer system that performs a dedicated function.

fetch-decode-execute cycle: the cycle through which data and instructions are processed.

architecture: the design of a computer system, including the components it contains.

3.1 The role of the CPU

The **CPU (central processing unit)** is one of the fundamental parts of a computer. Many people say it is the most important part, as it is where all the data and instructions are processed. This means it is a little bit like your brain, as that is where you process all the data and instructions that you sense on a daily basis.

All computers have a form of CPU. In devices such as personal computers (PC) and laptops, this is normally a separate component in the computer. In devices such as those found in our houses, e.g. our washing machine, a similar type of component exists called a **microprocessor**. This performs a similar role to a CPU, but on a smaller scale. Whereas a CPU can process thousands of complex instructions, a microprocessor will process a much more limited set of instructions. A general purpose computer, such as a PC or a laptop, has a CPU and is able to perform a wide variety of tasks. A microprocessor is often built into a system that is designed to perform a very specific task. This type of system is called an **embedded system**. There are examples of embedded systems in many different areas of our lives, for example, a traffic light, a washing machine and a digital alarm clock.

When an instruction is processed by a computer, the method that is used to do this is called the **fetch-decode-execute cycle**.

You need to understand what happens in the different stages of the cycle and which components are used. You'll learn some interesting information about how a computer works. How exciting!

You will learn the operation of the **architecture** of a Von Neumann computer system, which is one of the most common types of modern architecture for a computer system.

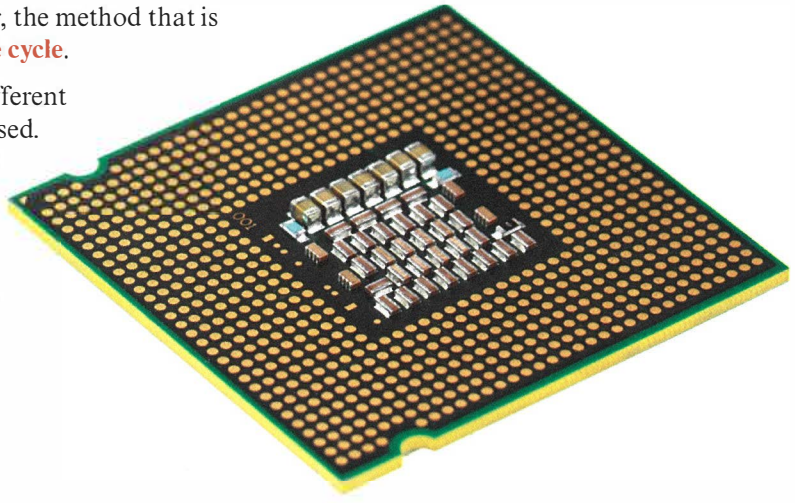


Figure 3.2: A CPU chip

ACTIVITY 3.1

Use the internet to research who Von Neumann was and why he developed the Von Neumann architecture.

This is a basic representation of the fetch-decode-execute cycle:

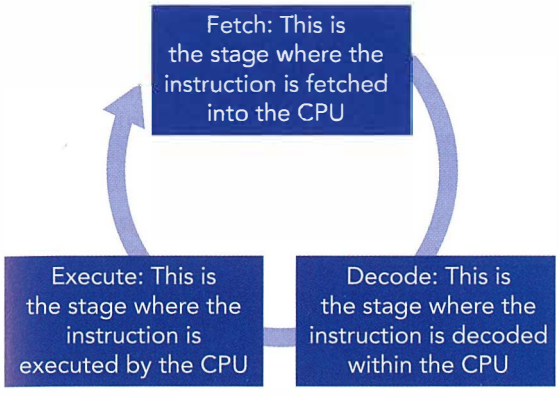


Figure 3.3: The fetch-decode-execute cycle

Individual processes happen at each of these stages, using different components, to carry out the cycle. So, what happens at each stage? We will explore this in the next sections.

3.2 The fetch stage

Data and instructions are given to the computer through the use of **input devices**. You will learn about these later in the chapter. When the data and instructions are input, they are stored in **RAM (random access memory)**. If a user wants to open a file that is stored on the **hard drive**, this data will be brought from the hard drive into the RAM, also.

In order for the data and instructions to be processed, they need to be brought from the RAM into the CPU. This is what happens at the fetch stage of the cycle. A component within the CPU, called the **program counter (PC)**, stores an **address**. This component is an example of a register. The address stored by the PC is the address of the next instruction that needs to be processed. The details of this address will be a location within the RAM.

This address stored in the PC needs to be sent to another component in the CPU called the **memory address register (MAR)**. This is another example of a register in the CPU. The address is transported from the PC to the MAR using the **address bus**.

Once the MAR has received the address it now knows where to find the instruction in the RAM. To retrieve the instruction the MAR needs to send the address to the location in RAM. It uses the address bus again to do this. Once the address is received at the location in RAM, the data stored in this location, e.g. the instruction, is sent to a

KEY WORDS

input device: a device that allows data to be entered into a computer system.

RAM (random access memory): a component in the CPU that holds data and programs that are currently in use.

hard drive: a storage device that stores data permanently, until it is deleted by the user.

program counter (PC): a component in the CPU that stores the address of the next instruction to be processed.

address: a memory location in RAM.

memory address register (MAR): a component in the CPU that holds the address of the data or instruction to be located in RAM.

address bus: wires used for the transmission of addresses between components in a computer.

component within the CPU called the **memory data register (MDR)**. This is also an example of a register in the CPU. The instruction is sent to the MDR using the **data bus**. When the MDR has received the instruction, it sends it to the **current instruction register (CIR)**. This is another example of a register in the CPU. The MDR uses the data bus to send the instruction to the CIR. This register is part of a component within the CPU called the **control unit (CU)**. It is the CU that is responsible for the next stage of the cycle, the decode stage.

Figure 3.4 shows the operation of the fetch stage of the cycle:

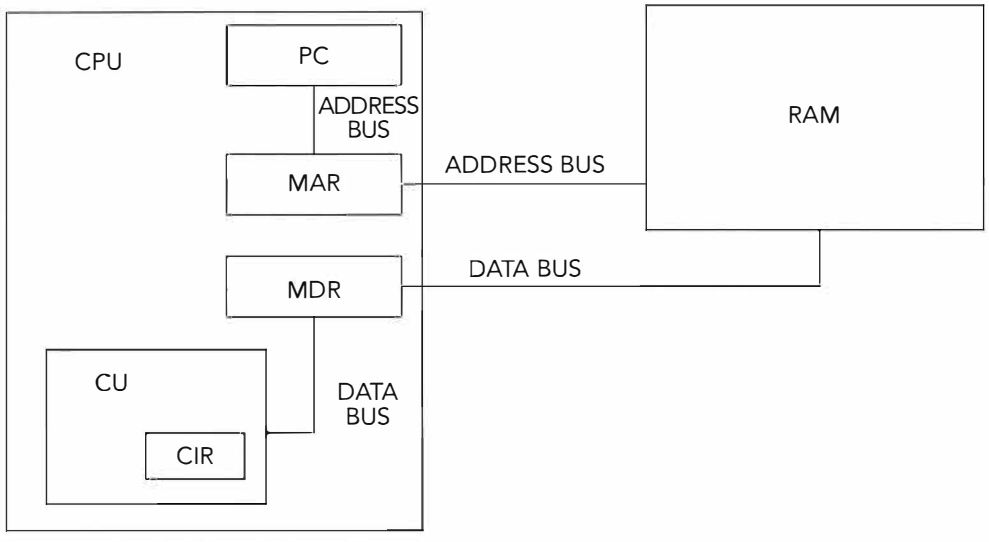


Figure 3.4: How the fetch stage operates

KEY WORDS

memory data register (MDR): a component in the CPU that holds the data or instruction that are fetched from RAM.

data bus: wires used for the transmission of data and instructions between components in a computer.

current instruction register (CIR): a register that is built into the CU that holds the current instruction that is being processed in the CPU.

control unit (CU): the component in the CPU that controls all the operations in the CPU.

instruction set: a set of commands that can be processed by a certain CPU.

ACTIVITY 3.2

Draw a diagram like the one given in Figure 3.4 to see if you can draw and label the process for the fetch stage of the cycle.

3.3 The decode stage

The decode stage of the cycle begins when the instruction is received by the CU. The CU is the unit in the CPU that decodes the instruction. The instruction needs to be decoded so that the CPU can understand what is required to execute the instruction. To decode an instruction the CU uses an **instruction set**. This is a set of all the commands that are understood and can be processed by the CPU. These commands are in machine code. Once the instruction is decoded using the instruction set, it can now be executed.

3.4 The execute stage

The execute stage is when any actions that are required for the instruction are carried out. This is likely to involve some mathematical and logical calculations. If these kinds

of calculations are required, the data used in the process is sent to the **arithmetic logic unit (ALU)**. This is a component within the CPU that is responsible for any calculations. Interim values may be created whilst these calculations are being performed, the ALU has a special register built into it called the **accumulator (ACC)**. It is this special register that will store the interim values that are created in any calculations. Once any calculations have been performed, the actions for the instruction can be performed, e.g. the file can be opened or the letter typed can be displayed.

How do all the different components know what they are doing in the fetch-decode-execute cycle? They have a conductor! This conductor is the CU. It sends signals to all the components within the CPU to tell them when they need to perform their different roles. The CU uses the **control bus** to send these signals.

KEY WORDS

arithmetic logic unit (ALU): the component in the CPU that performs all the mathematical and logical operations required when processing data and instructions.

accumulator (ACC): a register that is built into the ALU that stores the result of any interim calculations.

control bus: wires used for the transmission of control signals between components in a computer.

ACTIVITY 3.3

Add the ALU and the ACC to your diagram (from Activity 3.2) then label it to show the process for the decode and execute stages of the cycle.

Peer Assessment

Exchange diagrams from Activities 3.2 and 3.3 with a friend and use the description of the fetch-execute-decode cycle to check whether their diagram is correct.

Questions

1 Where is data stored before it is fetched into the CPU?

2 Which component in the CPU is responsible for making sure the fetch-decode-execute cycle is performed correctly?

3 What is the role of a bus?

4 Why does the ALU need the ACC? What would have to happen if it didn't have this special register?

5 Copy the table below and fill in a description for each component in the CPU:

Component	Description
Program counter (PC)	
Memory address register (MAR)	
Memory data register (MDR)	
Control unit (CU)	
Current instruction register (CIR)	
Arithmetic logic unit (ALU)	
Accumulator (ACC)	
Address bus	
Data bus	
Control bus	

3.5 Factors that can affect the performance of a CPU

The performance of a CPU refers to how quickly the CPU can process an instruction. This is important as the more instructions that can be processed in a second, the higher the performance of the CPU. Users often want a high performing CPU, so they are not sat waiting and frustrated with their computer because it takes too long to open a document, for example.

There are three main factors that can affect the performance of a CPU and this is the number of **cores** it contains, the **clock speed** and the size of the **cache**.

A single core in a CPU contains all the components that you have learnt that are used to perform the fetch-decode-execute cycle. Therefore, if a CPU has one core, it can perform one fetch-decode-execute cycle at a time. If a CPU has two cores (this is often called a dual core processor) it can perform two fetch-decode-execute cycles at the same time, one on each core. Some CPUs have four cores (this is often called a quad core processor) and can perform four fetch-decode-execute cycles all at the same time. Imagine completing a task: if you can only do one thing at a time to complete the task, it might take you quite a long time but if you could do four things at once to complete the task, you will complete it much quicker. Therefore, if a CPU has multiple cores, its performance can be increased.

Multiple cores also allow a computer to multitask. If it can process more than one instruction at a time, it can perform more than one task at a time.

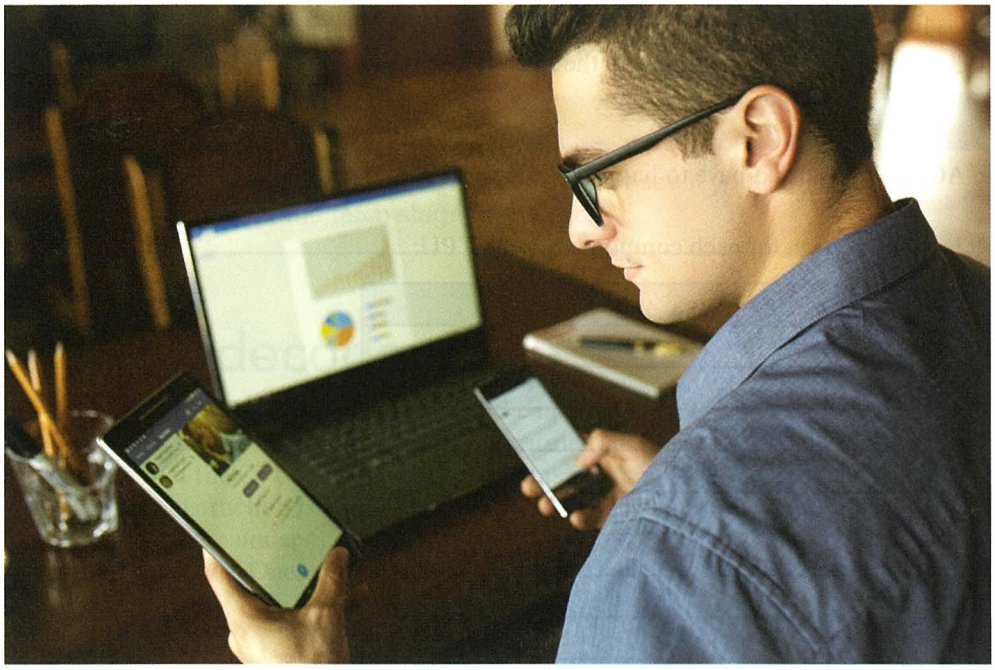


Figure 3.5: Multiple cores allow a computer to multitask

KEY WORDS

core: the part of the CPU that contains all the components that are used to perform the fetch-decode-execute cycle.

clock speed: the number of fetch-decode-execute cycles that can be performed in a second.

cache: a type of storage that is built into the CPU, to store the most frequently used data and instructions.

ACTIVITY 3.4

The performance of a CPU is not always increased by the addition of more cores. Use the internet to find out why the performance may not always be increased.

A CPU contains an internal clock that controls the speed at which instructions are processed. The speed is measured in hertz. A CPU that has a clock speed of 1 hertz is able to process one instruction per second. If your CPU had a clock speed of 1 hertz you'd be waiting rather a long time for any tasks to be completed!

Most modern computers have a clock speed that is measured in gigahertz (GHz). A CPU with a clock speed of 2 GHz can process 2 billion instructions a second. That is definitely a busy CPU! If a computer has two cores, each core can process 2 billion instructions a second.

ACTIVITY 3.5

It is possible to change the speed of a CPU using overclocking. Use the internet to find out what is meant by overclocking.

A CPU also contains a cache. The cache is a type of data storage within the CPU. Instructions and data that the computer uses most can be stored in the cache. This means that the CPU doesn't need to fetch this data from the CPU, just from the cache, so that process will be quicker. If the CPU can get the data and instructions quicker, the performance of processing those instructions will improve. If a computer has a larger cache, so it has more storage space to store the instructions and data it uses most, then it can store more of them, so will need to spend less time going to the RAM.

COMPUTER SCIENCE IN CONTEXT

Some people need very powerful processors in their computer, as they need to process a lot of data in a very short time period. One example would be an architect. They need to be able to use a computer to create a 3D model of their building design. It takes a lot of data to render these models, so a very powerful processor is needed to do this. 'Render' simply means to make the model. The computer does this by putting it together bit by bit.

Questions

6 Which CPU can process more instructions at the same time, a quad core or a dual core?

7 How many instructions per second can a dual core 2.4 Ghz CPU process?

8 If the cache size of a CPU is too big, this can actually have a negative effect on the performance of the CPU. Why might this happen?